

Q2/

$$\pi_{\text{firstName, lastName}} \left(\text{Person} \bowtie \text{Reservation} \right) \\ \text{Person.pid} = \text{Reservation.madeByPid}$$

Q3/

$$\pi_{\text{pid}} (\text{Attached}) \cup \pi_{\text{pid}} \left(\sigma_{\text{role} = 'PI'} (\text{Participates}) \right)$$

Q3/

$$\pi_{\text{labId}} \left(\sigma_{\text{building} = 'College of Computing'} (\text{Laboratory}) \right) \\ \cup \pi_{\text{labId}} \left(\pi_{\text{labId, modelId}} (\text{EquipmentUnit}) \bowtie \pi_{\text{modelId}} \left(\sigma_{\text{category} = 'CPU Server'} (\text{EquipmentModel}) \right) \right)$$

Q4/

$$A = \pi_{\text{labId, modelId}} (\text{EquipmentUnit})$$

$$\pi_{\text{labId}} \left(A \bowtie \pi_{\text{modelId}} \left(\sigma_{\text{category} = 'CPU Server'} (\text{EquipmentModel}) \right) \right)$$

$$\cap \pi_{\text{labId}} \left(A \bowtie \pi_{\text{modelId}} \left(\sigma_{\text{category} = 'GPU Server'} (\text{EquipmentModel}) \right) \right)$$

Q5/

$$\text{Supervisor} = \rho \left(\text{Supervisor} (\text{pid, labId}), \pi_{\text{supervisorPid, labId}} (\text{Laboratory}) \right)$$

$$CC = \pi_{\text{labId}} \left(\sigma_{\text{building} = 'College of Computing'} (\text{Laboratory}) \right)$$

$$\pi_{\text{pid, position}} \left(\text{Faculty} \bowtie (\text{Supervisor} \div CC) \right)$$

Q6/

$$\text{we have } \left\{ \begin{array}{l} A = \pi_{\text{pid, projectCode}} (\text{Participates}) \\ B = \pi_{\text{projectCode}} \left(\text{FundsProj} \bowtie \pi_{\text{budgetLine}} \left(\sigma_{\text{labId} = e} (\text{FundsLab}) \right) \right) \end{array} \right.$$

$$\hookrightarrow \pi_{\text{pid}} (A/B)$$

$$Q_7/ \quad C = p \left(C(pid, cnt), \gamma_{madeById} (count(resId) \rightarrow cnt) (Reservation) \right)$$

$$No_res = \pi_{pid} (Person) - \pi_{pid} (C)$$

$$Total = C \cup (No_res \times \{0\}) \quad \# \text{ cnt filled with 0 for no_res}$$

$$A = p(A(pid1, cnt1), Total)$$

$$B = p(B(pid2, cnt2), Total)$$

$$\pi_{pid1, pid2} (\sigma_{cnt1 > cnt2} (A \times B))$$

Q8/

$$R = \pi_{madeByPid, labId} (Reservation \bowtie Reserves \bowtie \pi_{serialNo, labId} (EquipmentUnit))$$

$$R_1 = p(R_1(pid, lab1), R)$$

$$R_2 = p(R_2(pid, lab2), R)$$

$$\pi_{pid} (\sigma_{lab1 \neq lab2} (R_1 \bowtie R_2))$$

$$Q_9/ \quad \left\{ \begin{array}{l} A = \pi_{resId} (\sigma_{submissionTS > '2026-09-01' \wedge submissionTS < '2026-10-01'} (Reservation)) \\ B = \sigma_{name = 'AI Movement'} (Laboratory) \end{array} \right.$$

$$\hookrightarrow \pi_{resId} (A \bowtie Reserves \bowtie EquipmentUnit \bowtie B)$$

Q10/

$$R = \pi_{pid, projectCode} (\sigma_{role = 'PI'} (Participates))$$

$$R_1 = p(R_1(pid, Proj1), R)$$

$$R_2 = p(R_2(pid, Proj2), R)$$

$$\hookrightarrow \pi_{pid} (\sigma_{Proj1 \neq Proj2} (R_1 \bowtie R_2))$$

Q₁₁/

$$A = \pi_{\text{approverPid, resId}} \left(\sigma_{\text{status} = 'approved' \wedge \text{approverPid} \neq \text{null}} (\text{reservation}) \right)$$

$$R = \pi_{\text{approverPid, labId}} \left(A \bowtie \text{Reserves} \bowtie \text{EquipmentUnit} \right)$$

$$R_1 = \rho(R_1(\text{pid, lab1}), R)$$

$$R_2 = \rho(R_2(\text{pid, lab2}), R)$$

$$\hookrightarrow \pi_{\text{pid}} \left(\sigma_{\text{lab1} \neq \text{lab2}} (R_1 \bowtie R_2) \right)$$

Q₁₂/

$$C = \pi_{\text{madeById}} \left(\sigma_{\text{PlannedStart} \leq '2026-11-06' \wedge '2026-11-06' \leq \text{PlannedEnd}} (\text{Reservations}) \right)$$

$$C_1 = \rho(C_1(\text{pid}), C)$$

$$\hookrightarrow \pi_{\text{pid}} (\text{Person}) - C_1$$

Q₁₃/

$$A = \gamma_{\text{labId}} \left(\text{counter} \leftarrow \text{count}(\text{serialNo}) \right) (\text{EquipmentUnit})$$

$$\text{No-unit} = \pi_{\text{labId}} (\text{Laboratory}) - \pi_{\text{labId}} (A) \quad \# \text{ Laboratory with no unit}$$

$$\text{Total} = A \cup (\text{No-unit} \times \{0\})$$

$$B = \gamma_{\text{labId}} \left(\text{avg} \leftarrow \text{AVG}(\text{counter}) \right) (\text{Total})$$

$$C = \pi_{\text{labId}} \left(\sigma_{\text{counter} < \text{avg}} (\text{Total} \bowtie B) \right)$$

$$\hookrightarrow \text{Laboratory} \bowtie C$$

Q14/

$$A = \pi_{\text{resId, madeByPid}} \left(\sigma_{\text{status} = \text{'Approved'}} (\text{Reservations}) \right)$$

$$B = \pi_{\text{fabId, resId, madeByPid}} (A \bowtie \text{Reserves} \bowtie \text{EquipmentUnit})$$

$$C = \gamma_{\text{fabId, madeByPid}} (\text{counter} \leftarrow \text{count}(\text{resId})) (B)$$

$$D = \gamma_{\text{fabId}} (\text{maximum} \leftarrow \text{max}(\text{counter})) (C)$$

$$\pi_{\text{fabId, madeByPid}} \left(\sigma_{\text{cnt} = \text{max}} (C \bowtie D) \right)$$

Q15/

$$A = \sigma_{\text{PlannedStart} \geq \text{'2025-01-01'} \wedge \text{PlannedEnd} < \text{'2026-01-01'}} (\text{Reservations})$$

$$B = \gamma_{\text{madeByPid}} (\text{counter} \leftarrow \text{count}(\text{resId})) (A)$$

$$\pi_{\text{madeByPid}} \left(\sigma_{\text{counter} \geq 3} (B) \right)$$